

Transient Stability Analysis and Multi-Stage UFLS Design for the Sumatra Interconnected Grid: A Comparative Case Study of the 2026 Blackout

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Abstract—The Sumatra interconnected power grid is a large longitudinal network spanning over 2,000 km, making it inherently vulnerable to cascading failures and system separation events. This paper presents a scenario-based transient stability analysis and Under-Frequency Load Shedding (UFLS) design for the N-2 double-circuit trip of the 275 kV Muara Bungo–Sungai Rumbai line during the May 22, 2026 Jambi Blackout. Using MATPOWER and ANDES, we conduct a controlled comparative study between the 2024 topology (without 500 kV backbone) and the 2026 topology (with the newly integrated 500 kV backbone) under identical load (3,397.22 MW). Results demonstrate that the 500 kV backbone reduces active transmission losses by 5.85% (from 93.19 MW to 87.74 MW) and improves the voltage profile. Sensitivity analysis shows that a minimum flat load shedding of 5% (≈ 107.5 MW) arrests the frequency decline in the 2026 deficit island, whereas the 2024 topology required 50% under a more severe contingency. We design a Wide-Area Measurement System (WAMS) assisted 3-stage UFLS scheme (5% at 49.5 Hz, 10% at 49.0 Hz, and 15% at 48.5 Hz) incorporating a 100 ms communication delay. Time-domain simulations confirm that this multi-stage scheme stabilizes the deficit island at 49.346 Hz by triggering only Stage 1 (5% load shed), shedding $4\times$ less load than a uniform 20% UFLS baseline while keeping frequency closer to nominal and preventing over-frequency violations.

Index Terms—Longitudinal grid, N-2 contingency, system islanding, transient stability, under-frequency load shedding (UFLS), WAMS.

I. INTRODUCTION

Modern power systems increasingly operate near physical limits due to rising load demands and variable renewable energy integration. Longitudinal grids—networks spanning long distances with limited parallel transmission paths—are particularly vulnerable to cascading failures during severe contingencies [1], [2]. Major blackout events worldwide highlight these vulnerabilities [3], [4]. The Sumatra interconnected grid in Indonesia exemplifies this challenge, stretching over 2,000 km from Aceh to Lampung and connecting southern generation centers to northern/central load centers through a narrow transmission corridor.

On May 22, 2026, extreme weather triggered the N-2 double-circuit trip of the 275 kV SUTET Muara Bungo–Sungai Rumbai line [5]–[7], severing the interconnection between southern

and north-central subsystems and causing frequency collapse across 8 provinces. A similar cascading failure occurred on June 4, 2024 on the Lahat–Lubuklinggau corridor [8]–[10]. Satellite data (TRMM-LIS) record flash densities exceeding 40 flashes/km²/year in central-south Sumatra [11], with BMKG reporting $> 60,000$ Cloud-to-Ground strikes monthly [12].

To enhance grid resilience, PT PLN (Persero) is integrating a 500 kV SUTET backbone (Muara Enim–New Aur Duri) under its 2025–2034 RUPTL [13]. While steady-state benefits are known, its quantitative impact on transient islanding dynamics and multi-stage UFLS coordination remains insufficiently investigated.

To bridge this research gap, this paper provides three primary contributions:

- 1) **Controlled Topology Comparison:** A rigorous comparative transient evaluation between the 2024 (without 500 kV backbone) and 2026 (with 500 kV backbone) configurations under identical load (3,397.22 MW) and dispatch. We note that each scenario uses its own historical N-2 contingency (Lahat–Lubuklinggau for 2024, Muara Bungo–Sungai Rumbai for 2026), so differences in shedding requirements reflect both topology and contingency severity.
- 2) **WAMS-Assisted Center-of-Inertia (COI) UFLS Modeling:** Application of classical machine dynamics and COI frequency tracking to model post-islanding frequency decay, incorporating a 100 ms WAMS communication and processing delay.
- 3) **Efficiency Metric & Validation:** Design of a 3-stage UFLS scheme and mathematical formulation of the Load Shedding Efficiency Ratio (LSER) metric to quantify frequency safety margin per MW shed while preventing over-frequency excursions.

The remainder of this paper is organized as follows. Section II details system modeling and formulation. Section III presents steady-state and islanding results. Section IV evaluates UFLS schemes and LSER efficiency. Section V concludes the paper.

II. SYSTEM MODELING AND FORMULATION

A. Power Flow Formulation

The steady-state behavior of the 521-bus Sumatra grid is solved using polar AC Newton–Raphson power flow in MATPOWER [14]:

$$P_i = V_i \sum_{j=1}^{N_b} V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) \quad (1)$$

$$Q_i = V_i \sum_{j=1}^{N_b} V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) \quad (2)$$

where V_i, V_j are voltage magnitudes, $\theta_{ij} = \theta_i - \theta_j$, and $G_{ij} + jB_{ij} \in Y_{\text{bus}}$ for $N_b = 521$ buses. Newton–Raphson mismatch tolerances are set to 10^{-6} p.u.

B. Generator Rotor Dynamics and Swing Equation

Time-domain transient stability simulations are conducted in ANDES [15]. Generator rotor dynamics use classical machine models (GENCLS) [16]:

$$\frac{d\delta_i}{dt} = \omega_i - \omega_0 \quad (3)$$

$$2H_i \frac{d\omega_i}{dt} = P_{mi} - P_{ei} - D_i (\omega_i - \omega_0) \quad (4)$$

where δ_i is rotor angle, ω_i is rotor speed, $\omega_0 = 2\pi \times 50$ rad/s, $H_i \in [3.0, 6.0]$ s is inertia constant, $D_i \in [1.5, 2.5]$ p.u. is damping, P_{mi} is mechanical power input, and P_{ei} is electrical power output. The integration time step is $\Delta t = 0.01$ s over a total duration of 30 s.

Upon system separation, the Center-of-Inertia (COI) frequency [16] characterizes overall subsystem dynamics:

$$f_{\text{COI}} = \frac{\sum_{i \in \mathcal{G}_{\text{is}}} H_i f_i}{\sum_{i \in \mathcal{G}_{\text{is}}} H_i} \quad (5)$$

where $\mathcal{G}_{\text{is}}$ is the generator set in the islanded subsystem and f_i is electrical frequency. Using f_{COI} filters out localized transient bus oscillations, preventing false relay tripping.

C. WAMS-Assisted Multi-Stage UFLS Model

Dynamic load shedding is modeled as a piecewise step function of f_{COI} [17]:

$$P_{d,j}(t) = P_{d0,j} \left(1 - \sum_{k=1}^{N_{\text{stage}}} d_k \cdot u(t - t_{\text{shed},k}) \right) \quad (6)$$

where $P_{d0,j}$ is pre-contingency bus load, d_k is stage shedding fraction, and $u(\cdot)$ is the Heaviside function. The trigger time $t_{\text{shed},k}$ incorporates Wide-Area Measurement System (WAMS) delay:

$$t_{\text{shed},k} = t(f_{\text{COI}} \leq f_{\text{thresh},k}) + t_{\text{delay}} \quad (7)$$

where $t_{\text{delay}} = 100$ ms accounts for PMU measurement (20 ms), fiber-optic transmission (50 ms), and relay/breaker operating time (30 ms) [18], [19]. The proposed 3-stage UFLS parameters are: Stage 1 ($f_{\text{thresh},1} = 49.5$ Hz, $d_1 = 0.05$), Stage 2 ($f_{\text{thresh},2} = 49.0$ Hz, $d_2 = 0.10$), and Stage 3

($f_{\text{thresh},3} = 48.5$ Hz, $d_3 = 0.15$) [20], [21]. Stage 1 aligns with the lower bound of the normal deadband (49.5 Hz) to initiate early mitigation.

D. Contingency Selection Rationale

The Muara Bungo–Sungai Rumbai corridor is selected based on failure probability $P_i(T) = 1 - e^{-\lambda_i T} \approx \lambda_i T$ under a Poisson process [4], where $\lambda_i \propto X_i$ (line reactance as proxy for length and lightning exposure). Calibrated with BMKG lightning data ($\lambda_{\text{base}} = 0.01/\text{year}$), this 275 kV double circuit represents the highest-risk N-2 corridor. Simulations evaluate deterministic N-2 line opening ($P = 1$) at $t = 1.1$ s (100 ms post fault at $t = 1.0$ s).

III. STEADY-STATE AND ISLANDING RESULTS

A. Grid Base Case Comparison

Steady-state parameters for 2024 and 2026 grid models under identical baseline load ($P_{\text{load}} = 3,397.22$ MW) are summarized in Table I. The 2026 model includes the 500 kV SUTET backbone (Muara Enim–New Aur Duri–Peranap–Perawang), adding 5 buses (including Bus 439 PLTU Sumsel-8 and Bus 436 PLTU Bengkulu 500 kV) and 2 generator units (installed capacity 9,726.51 MW vs. 9,591.71 MW).

TABLE I
STEADY-STATE PARAMETER COMPARISON (IDENTICAL LOAD: 3,397.22 MW)

Parameter	2024 Grid	2026 Grid
Active Buses / Generators	516 / 68	521 / 70
Installed Capacity (MW)	9,591.71	9,726.51
Active Load / Generation (MW)	3,397.22 / 3,490.41	3,397.22 / 3,484.95
Transmission Losses (MW)	93.19 (2.67%)	87.74 (2.52%)
Min / Max Voltage (p.u.)	0.9481 / 1.0639	0.9547 / 1.0500

The 500 kV backbone reduces active losses by 5.85% (from 93.19 MW to 87.74 MW) and elevates minimum voltage from 0.9481 p.u. to 0.9547 p.u. The 2024 maximum voltage of 1.0639 p.u. occurs at lightly loaded 275 kV buses in Central Sumatra, remaining within the grid code upper limit of 1.07 p.u.

B. 2024 Grid: Historical Blackout Reconstruction

In the 2024 grid without 500 kV backbone redundancy, N-2 trip of the Lahat–Lubuklinggau 275 kV line together with its parallel 150 kV Prabumulih–Simpang Tiga circuit at $t = 1.1$ s severs the primary north-south interface. Unmitigated, progressive reactive power mismatch causes dynamic voltage decay, leading to numerical solver divergence at $t \approx 4.3$ s (Fig. 1). This divergence reflects loss of transient rotor angle/voltage stability. Applying flat 50% UFLS ($\approx 1,075$ MW shed at $t = 1.2$ s) arrests collapse but causes severe over-frequency (52.416 Hz), exceeding the 51.5 Hz OFLS limit and confirming that uniform UFLS without backbone redundancy causes secondary trip risks.

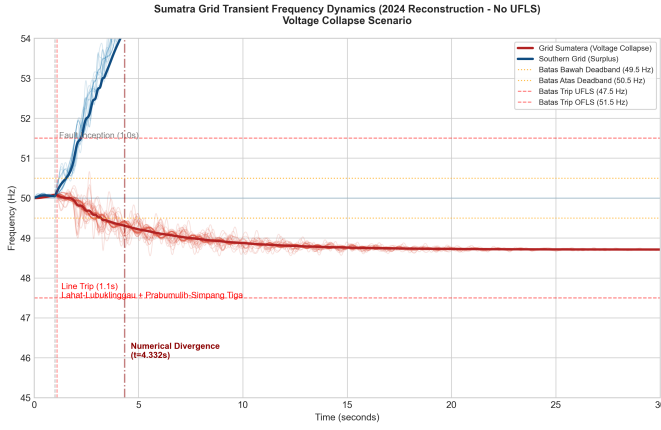


Fig. 1. Dynamic frequency decay in the 2024 grid following N-2 trip of Lahat–Lubuklinggau line, leading to numerical divergence at $t \approx 4.3$ s.

C. 2026 Scenario: N-2 Islanding Dynamics

Upon N-2 trip of the 275 kV Muara Bungo–Sungai Rumbai line at $t = 1.1$ s, the 2026 grid splits into two isolated subsystems: (1) **Southern Island** (226 buses, generation surplus, frequency rises slightly); and (2) **North-Central Island** (294 buses, active power deficit of 112.68 MW, steep frequency decay).

Without UFLS, the north-central island frequency declines steadily from 50 Hz to 48.79 Hz at $t = 30$ s (Fig. 2a), remaining on a collapse trajectory toward the 47.5 Hz generator under-frequency trip threshold. Fig. 2 shows the response for 0% and 20% flat UFLS.

IV. UFLS PERFORMANCE AND MULTI-STAGE VALIDATION

A. Uniform Load Shedding Sensitivity

Flat block load shedding sensitivity at $t = 2.0$ s under identical conditions is presented in Table II.

TABLE II
UFLS SENSITIVITY RESULTS FOR 2026 GRID (DEFICIT: 112.68 MW)

UFLS (%)	f_{final} (Hz)	f_{nadir} (Hz)	t_{nadir} (s)	RoCoF (Hz/s)	Stable	OFLS Trip
0	48.79*	48.79	30.0*	0.31	No	No
5	49.35*	49.35	30.0*	0.31	Yes	No
10	49.93	49.77	2.00	0.31	Yes	No
15	50.52	49.77	2.00	0.31	Yes	No
20	51.14	49.77	2.00	0.35	Yes	No
25	51.77	49.77	2.00	0.51	Yes	Yes
30	52.43	49.77	2.00	0.66	Yes	Yes
35	53.11	49.77	2.00	0.83	Yes	Yes
40	53.80	49.77	2.00	0.99	Yes	Yes

*For 0% and 5%, the frequency is still declining at $t = 30$ s (nadir at the end of the simulation window); for UFLS $\geq 10\%$, the nadir occurs at $t = 2.0$ s before recovery.

Flat shedding of 5% (≈ 107.5 MW) arrests collapse and stabilizes frequency at 49.35 Hz, which rests slightly below the normal deadband (49.5 Hz) but safely above the critical generator trip limit (47.5 Hz). Flat 10% shedding stabilizes frequency at 49.93 Hz inside the normal deadband. However, UFLS $\geq 25\%$ causes over-frequency exceeding the 51.5 Hz OFLS limit. For UFLS $\geq 10\%$, the frequency nadir (49.77 Hz at $t = 2.00$ s) is identical because the flat block is applied at

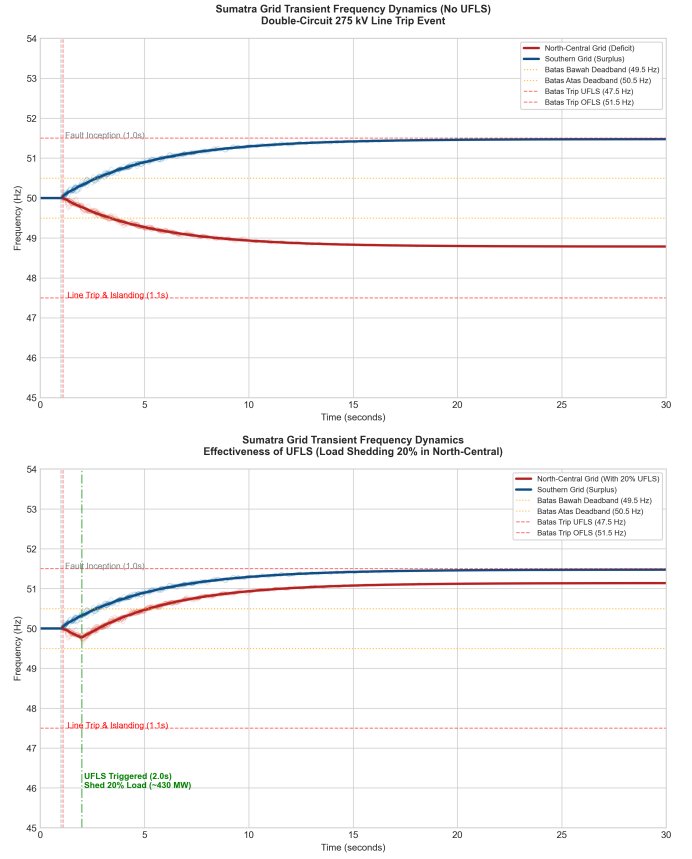


Fig. 2. Transient frequency response of 2026 Sumatra grid under N-2 blackout. (a) 0% UFLS: frequency declines steadily toward the 47.5 Hz trip limit (48.79 Hz at $t = 30$ s). (b) 20% flat UFLS at $t = 2.0$ s: frequency stabilized at 51.14 Hz. Dashed orange: deadband (49.5–50.5 Hz); dashed red: UFLS (47.5 Hz) and OFLS (51.5 Hz) trip limits.

$t = 2.0$ s and the nadir is governed by pre-UFLS inertia and deficit.

B. Multi-Stage UFLS Validation

The proposed 3-stage scheme operates as follows: (1) **Stage 1 Trigger:** at $t = 3.38$ s, f_{COI} crosses 49.5 Hz, and the 5% load shed (≈ 107.5 MW) is applied at $t = 3.48$ s after the 100 ms WAMS delay; (2) **Stabilization:** the 5% shedding arrests the frequency decline, stabilizing grid frequency at 49.346 Hz; (3) **Selective Operation:** because frequency remains above 49.0 Hz, Stages 2 and 3 remain untriggered.

Post-mitigation steady-state frequency of 49.346 Hz is an acceptable temporary emergency state above 47.5 Hz, preserving grid integrity until secondary frequency control (governor/operator re-dispatch) restores 50.0 Hz. Fig. 3 illustrates the dynamic response.

C. Load Shedding Efficiency Ratio (LSER)

To quantitatively compare UFLS efficiency, we formulate the Load Shedding Efficiency Ratio (LSER) in Hz/MW:

$$\text{LSER} = \frac{f_{\text{stable}} - f_{\text{trip}}}{P_{\text{shed}}} \quad (8)$$

where f_{stable} is stable post-mitigation frequency, $f_{\text{trip}} = 47.5$ Hz is the critical generator under-frequency trip limit, and P_{shed} is total active power shed in MW. LSER quantifies the frequency safety margin secured per MW of shed load.

Applying Eq. (8) across the flat schemes reveals diminishing returns as shedding increases:

- **Flat 5%** (107.5 MW, 49.35 Hz): $\text{LSER} = (49.35 - 47.5)/107.5 \approx 0.0172$ Hz/MW.
- **Flat 10%** (215 MW, 49.93 Hz): $\text{LSER} = (49.93 - 47.5)/215 \approx 0.0113$ Hz/MW.
- **Flat 20%** (430 MW, 51.14 Hz): $\text{LSER} = (51.14 - 47.5)/430 \approx 0.0085$ Hz/MW.

The monotonic decline in LSER quantifies the cost of over-shedding: each additional MW of shed load secures progressively less frequency margin, and beyond 20% the scheme risks over-frequency trips. The **multi-stage scheme** (107.5 MW, 49.346 Hz) achieves $\text{LSER} = (49.346 - 47.5)/107.5 \approx 0.0172$ Hz/MW, matching the efficiency of the most economical flat scheme (5%) while adding adaptivity: it sheds the minimum load required by the actual deficit, and Stages 2–3 would shed more only if the frequency continued to decline. Relative to the flat 20% baseline, the multi-stage scheme sheds $4\times$ less load (107.5 vs. 430 MW) while keeping frequency closer to nominal (49.35 vs. 51.14 Hz)—yielding an LSER $2.03\times$ higher and eliminating over-frequency trip risks.

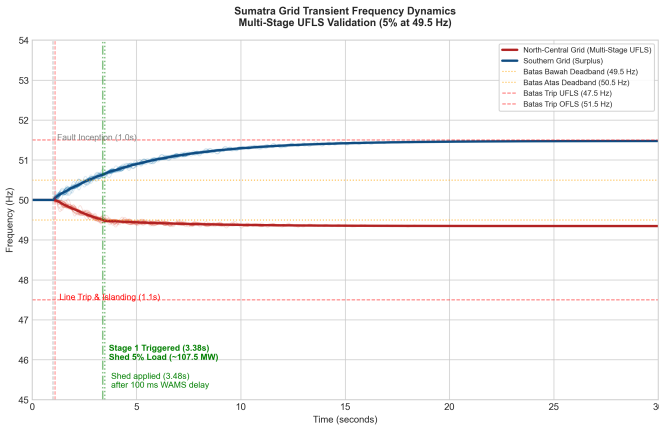


Fig. 3. Transient frequency dynamics of 3-stage UFLS scheme for 2026 Sumatra grid. Only Stage 1 (5% at 49.5 Hz) triggers at $t = 3.38$ s, stabilizing frequency at 49.346 Hz without activating Stages 2 and 3.

V. CONCLUSION

This paper presents a transient stability analysis of the Sumatra grid under N-2 islanding events, comparing 2024 (without 500 kV backbone) and 2026 (with backbone) configurations. Key findings are:

- 1) The 500 kV backbone reduces transmission losses by 5.85% (from 93.19 MW to 87.74 MW) and improves the minimum voltage from 0.9481 to 0.9547 p.u. The 2026 deficit island requires only 5% flat load shedding to arrest the frequency decline, whereas the 2024 topology required 50% under its more severe Lahat–Lubuklinggau

contingency—a difference reflecting both backbone redundancy and contingency severity.

- 2) Without backbone redundancy, 50% UFLS in 2024 causes 52.416 Hz over-frequency (exceeding the 51.5 Hz OFLS limit), confirming that a uniform shedding strategy without a low-impedance bypass is prone to secondary tripping.
- 3) The WAMS-assisted 3-stage UFLS scheme stabilizes the deficit island at 49.346 Hz with only 5% load shed (107.5 MW). It matches the LSER of the most economical flat scheme (5%) while shedding $4\times$ less load than a uniform 20% scheme, keeping frequency closer to nominal and eliminating over-frequency risks.

Limitations & Screening Notice: Classical machine models (GENCLS), constant impedance loads, and simplified control representations serve as an initial screening tool. Field implementation requires detailed turbine-governor, AVR, composite ZIP load modeling, WAMS integration, and Hardware-in-the-Loop (HIL) testing [18]–[21].

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DATA AND CODE AVAILABILITY

Sumatra 521-bus grid models (MATPOWER .m and ANDES .xlsx), Python simulation scripts, and data analysis codes are open-access at: <https://gitlab.com/novalio.daratha/blackout-s-umatera>.

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